

Metachronous Brain Metastases in NSCLC Show Markedly Worse Post-Metastasis Survival Despite a Shared Genomic Landscape

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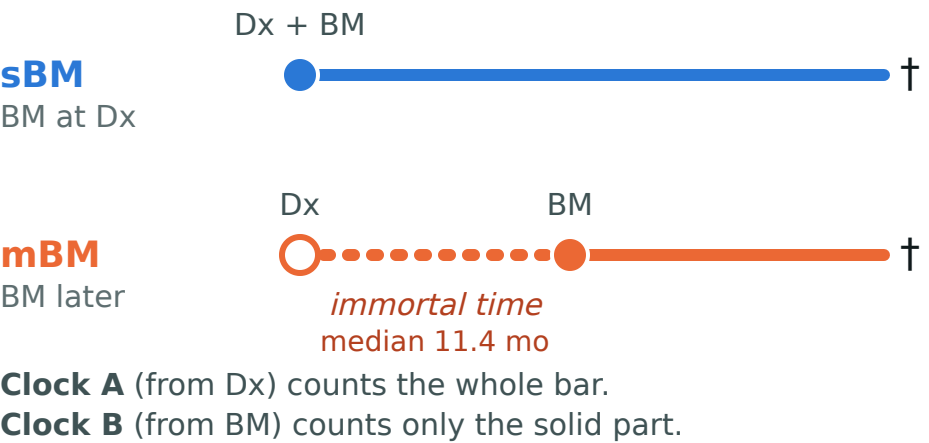
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BACKGROUND

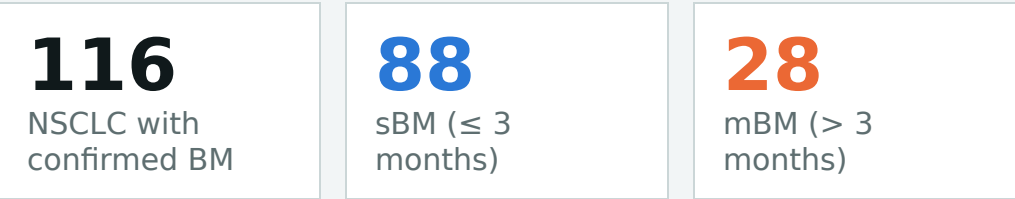
Brain metastases (BM) develop in up to **40% of NSCLC**. Whether **synchronous** (at diagnosis, sBM) and **metachronous** (later, mBM) BM differ in prognosis or genomics is unclear.

WHY THE SURVIVAL CLOCK MATTERS

Survival is usually measured from primary diagnosis, but an mBM patient must survive until BM appears: **immortal time**.



METHODS

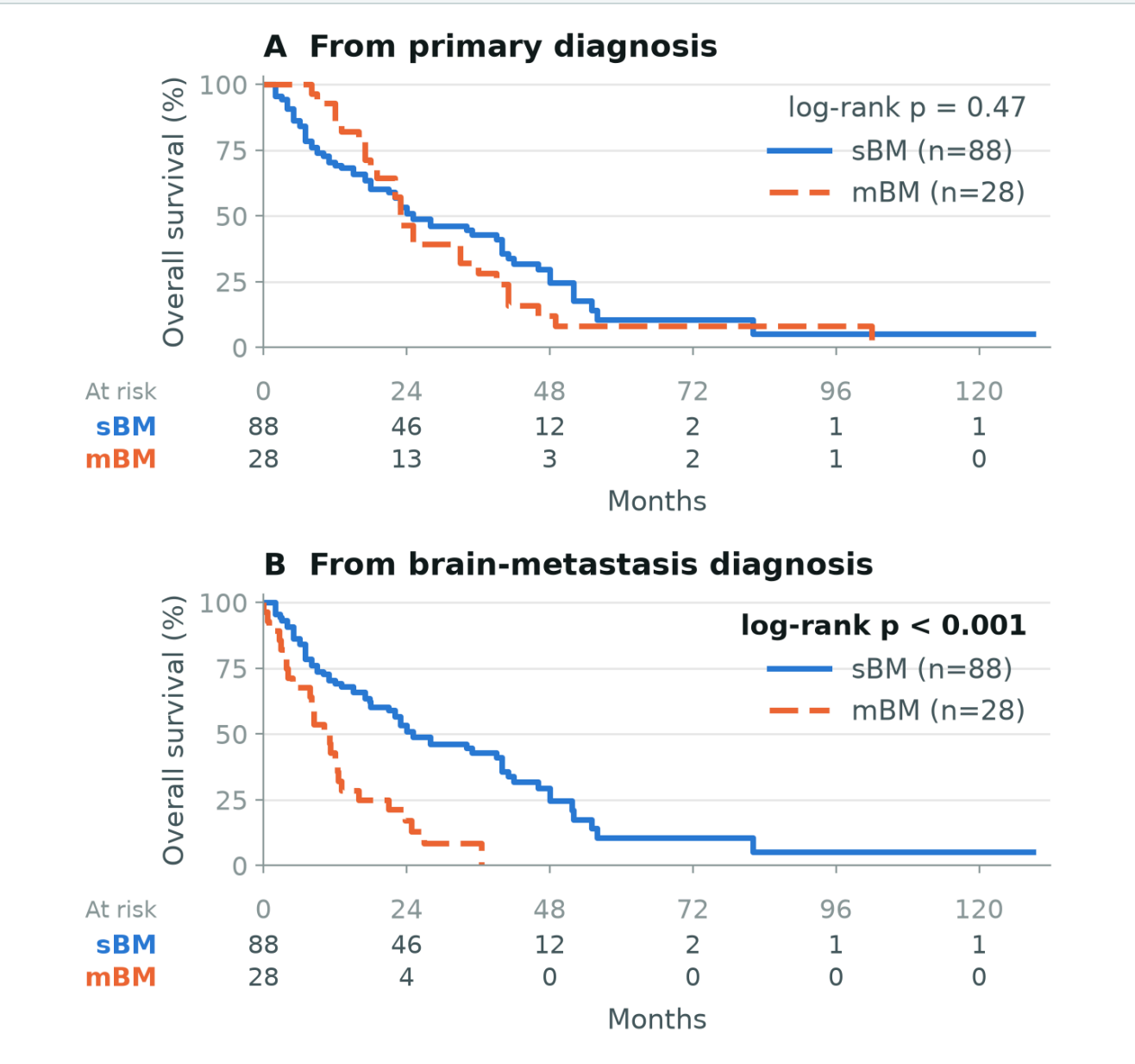


- Monocentric, retrospective, consecutive; 89 deaths. Balanced for age, sex, histology.
- FoundationOne CDx genomic profiling; PD-L1 by SP263.
- Kaplan-Meier, log-rank, Cox. 42 recurrent alterations (≥ 5 patients): Fisher's exact test, Benjamini-Hochberg FDR.

ENDPOINTS

Primary: OS from BM diagnosis (clock B)
Secondary: OS from primary diagnosis (clock A)

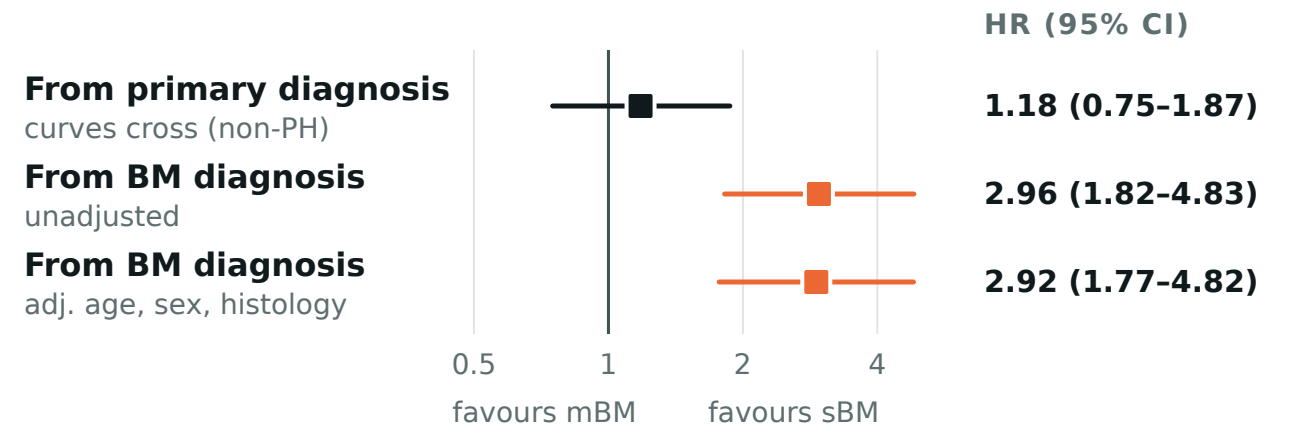
RESULTS · SURVIVAL



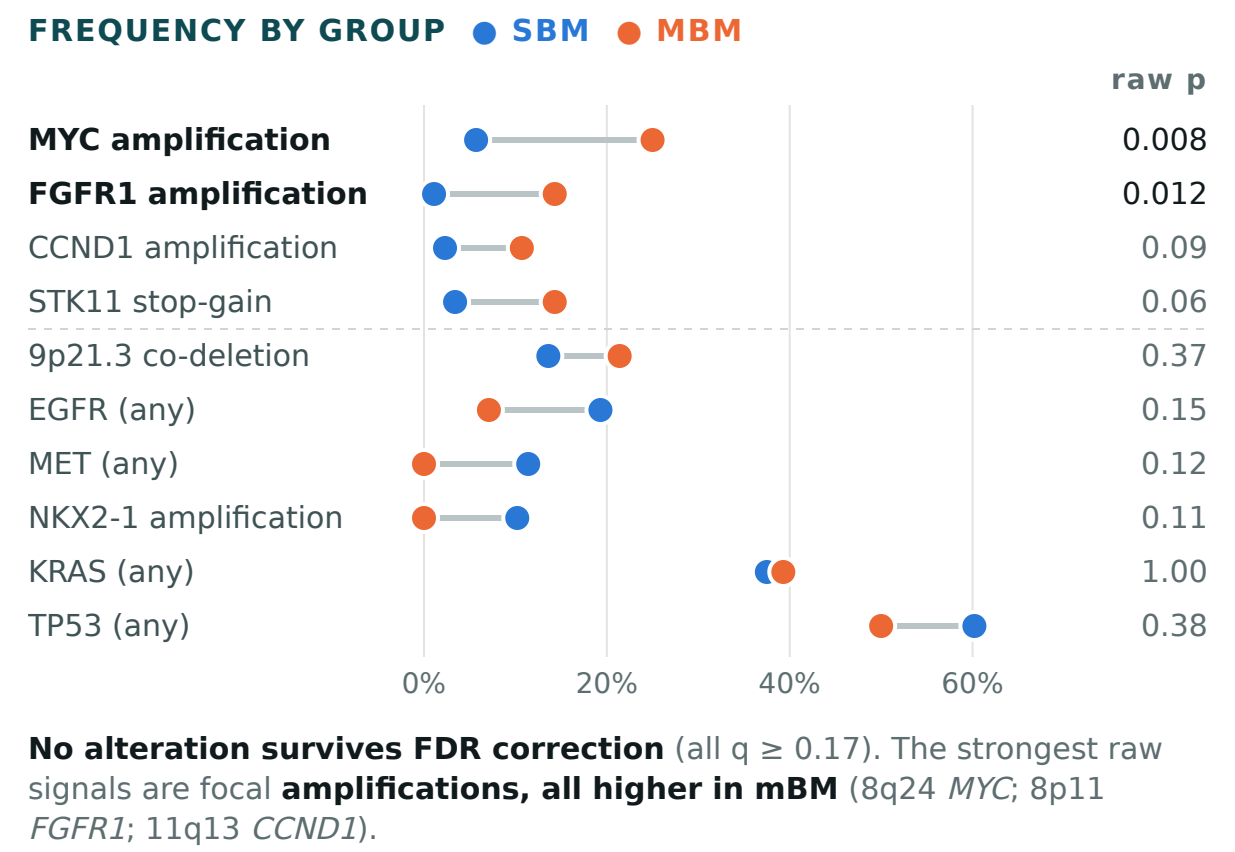
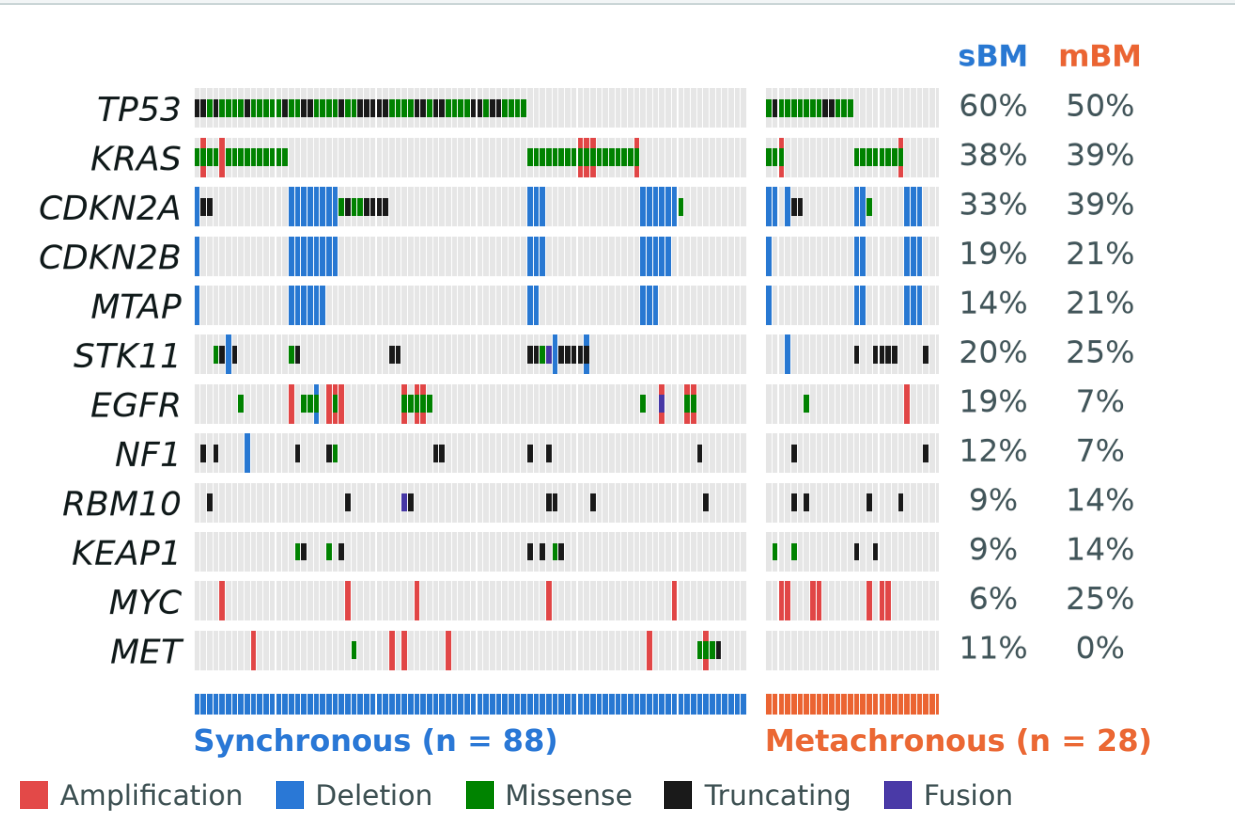
Same 116 patients, 89 deaths. In A, mBM runs *above* sBM early (it cannot die before BM), then below: the curves cross. Median OS from BM, 95% CI: sBM 18.0–40.0, mBM 4.9–13.1 mo.



MBM VS SBM · COX MODELS



RESULTS · GENOMICS



9P21.3 CDKN2A/CDKN2B/MTAP CO-DELETION
15.5% (18/116; sBM 13.6% vs mBM 21.4%, p = 0.37), always as one contiguous deletion. Not above unselected NSCLC on comparable CGP (13.4%, n = 29,379; p = 0.49). No detectable prognostic effect: HR 1.17 (0.65–2.10).

INTERPRETATION

- The clock decides the answer.** From primary diagnosis, immortal time and later excess mortality cancel out. From BM diagnosis, mBM carries **about 3x the hazard**, stable under adjustment.
- mBM is a **progression event in pretreated disease**; its survival resembles later-line rather than first-line outcomes.
- No genomic difference detected, but not equivalence:** 28 mBM patients detect only ~4-fold differences (e.g. 15% vs 43%). The mBM *MYC* amplification excess echoes BM-enriched *MYC* amplification (Shih et al.¹) and merits validation.
- 1 in 6 carries an MTAP deletion**, independent of BM timing: candidates for PRMT5/MAT2A inhibitor trials.

CONCLUSION

Survival after metachronous BM is short. Compare sBM and mBM from the metastatic event: from primary diagnosis, the difference disappears.

For counselling and trial stratification, treat mBM as disease that has progressed on therapy.

LIMITATIONS

Single centre, retrospective, 28 mBM. Annotation missing unevenly (43% sBM vs 21% mBM); in the 72 fully annotated patients HR 1.85 (1.07–3.21). CNS-directed therapy not captured. No BM-free comparator, so no claim about BM *risk*.

NEXT STEPS

Add a BM-free CGP comparator (enrichment, time-dependent Cox). Abstract SRS/WBRT/resection per patient. Validate the mBM amplification signal in an external series.

¹ Shih DJH et al. *Nat Genet* 2020;52:371–377. Reference 9p21.3 rate: Kumar et al., *Cancer Med* 2023.